

GRANDlib: merged, cleaned, validated, beta- tested

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VILLUM FONDEN



Since the last update

	What	Status
Branches	All 40 branches decided; dev-next holds everything kept	done
Old PRs and issues	8 pull requests closed, 1 ported; 12 issues closed in triage, 7 fixed	done
Input validation	Bad input refused with a clear message: 50 of 60 cases, up from 7	merged, PR #179
Beta test	Two waves, 19 testers, 91 issues; all 11 Critical issues fixed	PR #208, CI green

Cleaning up

The recovery plan: 65 of 90 items done

dev-next recovery, by phase

65 of 90 items done across 11 phases. Read from the plan's own checkboxes; every branch and its status is in branches.svg.

☒ done ☐ in progress ☐ blocked on a decision ☐ not started

0 integration branch

3 of 4 done

next: Announce the freeze date for dev

1 one environment

4 of 5 done

next: Verify the four-command install on...

✓ 2 restore CI

all 8 done

0 3 tests first

7 of 8 done

next: External check against a published...

✓ 4 merge the queue

all 11 done

× 5 the decisions

5 of 6 done

next: Docker: publish an image, or state...

0 6 delineate I/O

0 of 5 done

next: Extract the pure kernel — arrays...

✓ 7 documentation

all 19 done

0 8 governance

5 of 8 done

next: Move large ROOT fixtures to a...

0 9 promote to main

1 of 4 done

next: Archive the 2023 main, free the name

0 10 bulk cleanup

2 of 12 done

next: Re-run git cherry against the...

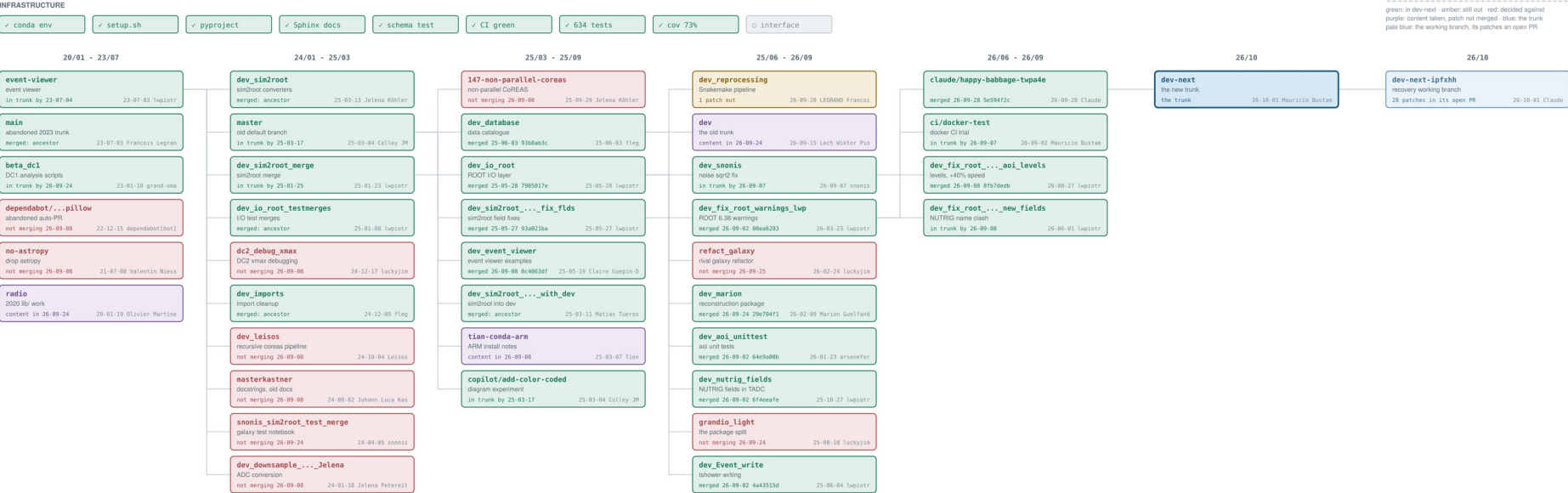
BLOCKED ON A DECISION

- Docker: publish an image on ROOT 6.36, or state that Docker is unsupported — needs the collaboration
- Reprocessing: the noise fix raises every simulated voltage by $\sqrt{2}$ — needs the collaboration

Every branch decided: 40 branches

Every branch in grand and its status

40 branches, joined to their nearest ancestor. Columns are generations, oldest on the left.

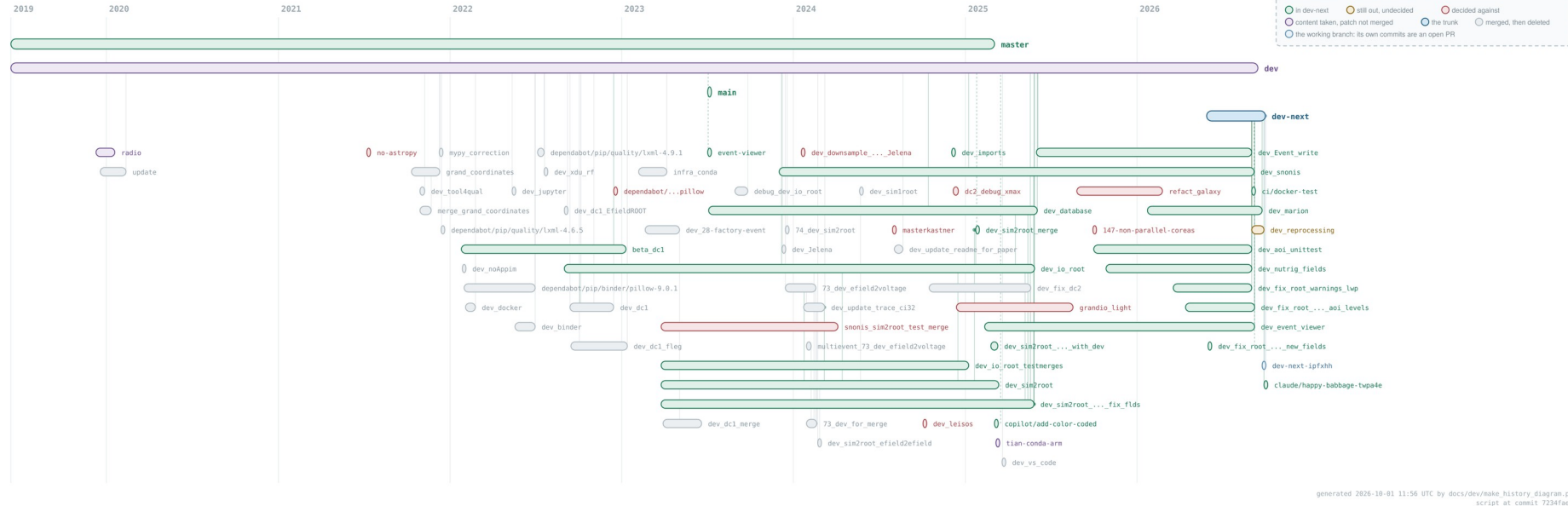


generated 2026-10-01 11:55 UTC by docs/dev/make_branch_diagram.py
script at commit 7234fa3

Seven years of branches, 2019 to 2026

Every branch and merge in grand, 2019 to 2026

71 branches over 7 years, oldest at the top. Context for the recovery rather than part of it: most of these no longer exist.



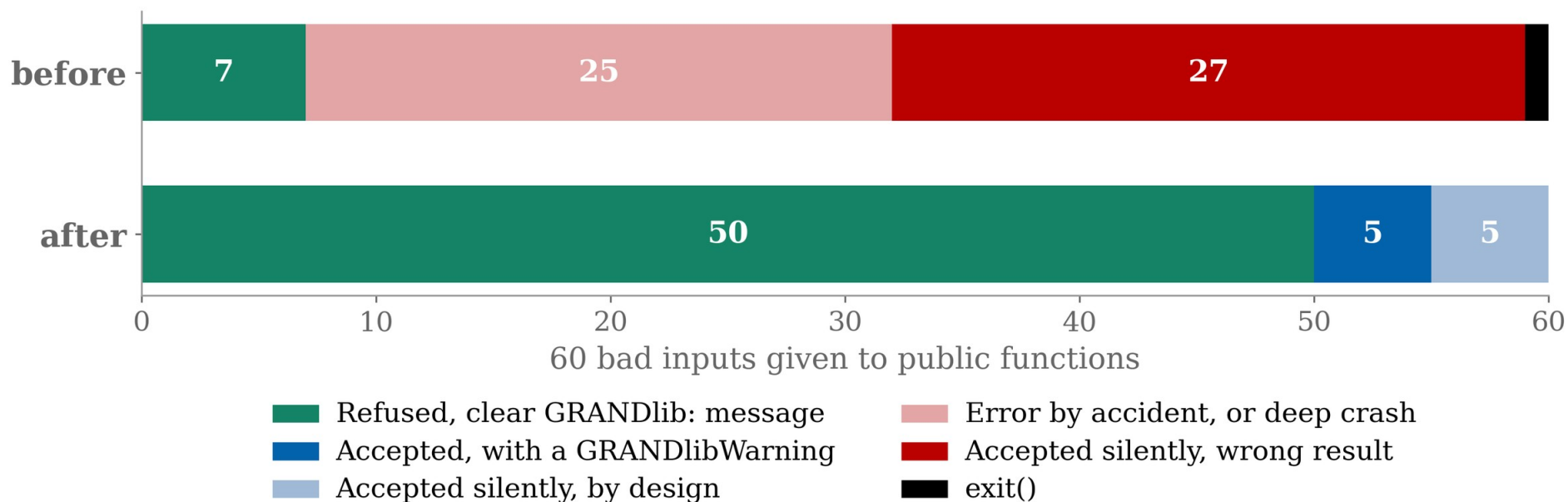
Old pull requests and issues

	Count	What happened
Pull requests	8 + 1	Closed: #9, #49, #52, #146, #149, #151, #153, #154. Ported by hand: #150 (Cramér–Rao bounds)
Old issues, triaged	12	Fixed (#95, #122, #123, #136); duplicate (#80, #84, #94, #99); not reproducible (#90, #92, #156); obsolete (#47)
Old issues, fixed	7	#71, #89, #91, #104 (Xmax, direction), #137, #139 (T1 trigger), #140 (reader)
Left for owners	4	#85, #121, #141, #142

Input validation

Bad input now stops with a clear message

60 bad inputs given to public functions: wrong type, range, shape or file. **Before:** 27 gave a wrong result silently. **After:** none do.



The beta test

The beta test: three waves

Testers use GRANDlib as a newcomer, an expert or a pipeline user would, and report only.

Every report is reproduced before it becomes a GitHub issue, titled *dev-next_beta-test: ...*, with its severity.

Fixes go into one pull request, one commit and one test each. A test must fail without its fix.

Wave	Testers	Looks for	Issues
1. Using it	11	Missing steps, wrong docs, brittle input	69
2. Checking it	8	Physics, docs, error handling, tests	22
3. Breaking it	5	Misuse, extremes, stress, unsafe input	next

Wave 1: using it

Tester	Area	Issues	Critical	High
Beginners (×3)	Setup, the 12 notebooks, examples	14	–	2
Expert, dataio	Every tree, reader and writer	12	–	5
Expert, analysis	aoi, reconstruction, event viewer	7	1	3
Pipeline, ZHAireS	ZHAireS → rawroot → sim2root	8	1	3
Pipeline, CoREAS	CoREAS, e-field → voltage → ADC	9	4	2
Input fuzzers (×4)	Readers, scripts, damaged simulations	18	6	8

69 issues: **10 Critical**, 22 High, 26 Medium, 11 Low. Most Critical ones were in the conversion chain.

Wave 2: checking it

Tester	Area	Issues	Critical	High
Physics (×2)	Coordinates, Xmax, signal chain, noise	7	1	2
Documentation (×2)	Docs, Handbook, READMEs, docstrings	4	–	1
Error handling	Silent failures, prints, asserts	3	–	2
Test suite	Coverage, deliberate bugs, weak tests	2	–	1
Input validation	Type, range, shape, units	6	–	3
Notebook prose	Every claim against its output	2	–	1

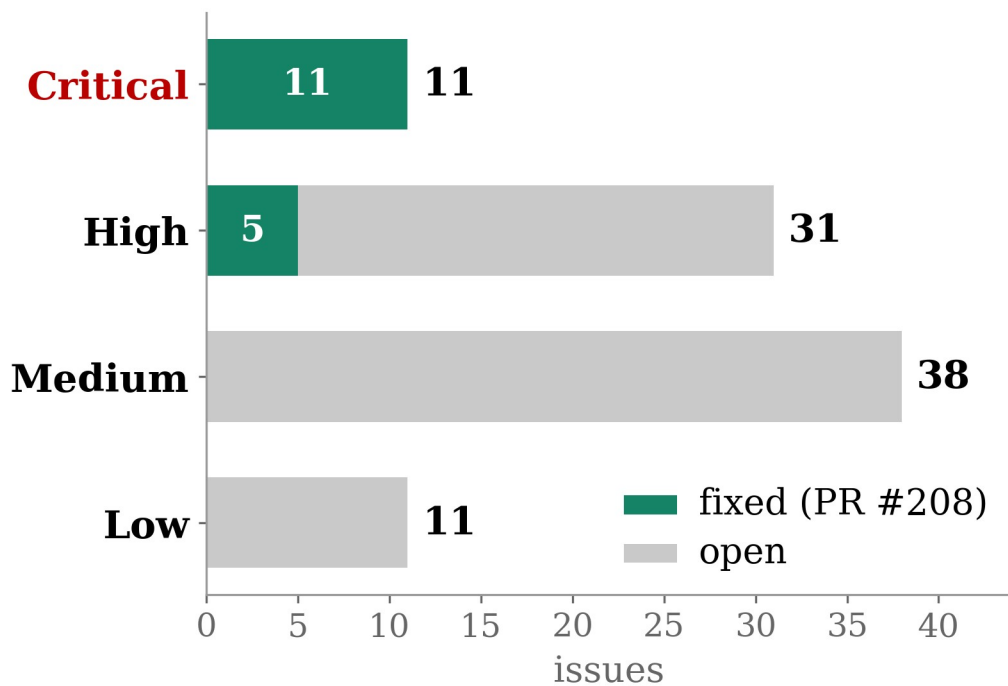
22 issues: **1 Critical**, the EGM96 geoid upside down (−7.75 m at GP300 instead of −61.0 m). The test suite let 5 of 15 deliberate bugs through.

Wave 3: breaking it

Breaker	Attack
Misuse	Methods in the wrong order; one object across files; events that do not exist
Numerical edges	Zenith 0° and 90°; NaN through a whole chain; one-antenna events
Scale and stress	Thousands of files; long traces; memory over long loops
Environment	Unset variables; missing packages; read-only or partial data
Unsafe input	File names that reach a shell; archive extraction

Starts once PR #208 is merged. A regression wave then re-runs the busiest testers on the fixed code.

91 issues; every Critical one fixed



Severity	Issues	Fixed	Open
Critical	11	11	0
High	31	5	26
Medium	38	0	38
Low	11	0	11
Total	91	16	75

Critical: wrong physics or data, silently.

High: crashes, data loss, documented commands that fail.

The 11 Critical bugs, now fixed

Issue	Was wrong	Effect
#209	CoREAS azimuth mirrored	Wrong arrival direction
#220	sim2root antenna latitude and longitude	Off by up to 8 km
#227, #229	RF-chain options ignored; resampled rate lost	Wrong voltages and ADC rate
#228	Unknown CoREAS Xmax read as -1 cm	Voltages near 10^{-13} μ V
#237	L0 e-field paired with an L1 run tree	Amplitudes doubled
#238, #247	A missing event or shower	Another event's data
#242, #243	Damaged simulations accepted	Valid-looking files
#250	EGM96 geoid upside down	Heights off by 53 m at GP300

Still to fix

Area	Open High issues
Data loss	<code>write()</code> replaces trees; <code>Event.write(overwrite=True)</code> ; <code>extract_events -ow</code> ; re-runs; <code>get_files_from_db.py</code>
Silent wrong results	<code>DataDirectory</code> drops files; event order; NaN in the ADC; geodetic heights west of Greenwich
Commands that fail	README quickstart; pipeline commands; docs examples; helper scripts
Crashes and hangs	<code>close_files()</code> ; <code>EventList</code> options; the <code>rf_chain</code> error path

Next: the 26 High issues, data loss first. The 38 Medium and 11 Low are mostly docs and messages.

What we need from you

Access to the GRAND data at CC-IN2P3 (SPS), to run the aoi part of GRANDlib on real events.

The aoi readers and the reconstruction scripts expect the data under `/sps/grand/data/`. So far they have been tested only on simulations and on the small files committed to the repository.

An account, or a few event files copied out, would be enough.

Timeline

